

ADVANCED

Stage 6

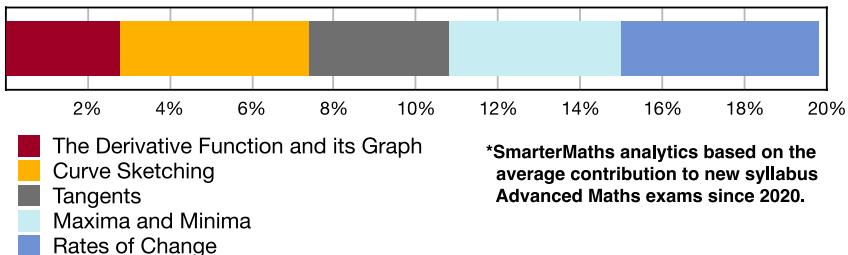
Calculus (Adv), C3 Applications of Calculus (Adv)

Rates of Change (Y12)

Teacher: SHS

Exam Equivalent Time: 87 minutes (based on allocation of 1.5 minutes per mark)

C3 Applications of Calculus



HISTORICAL CONTRIBUTION

- C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: 1-*The Derivative Function and its Graph* (2.8%), 2-*Curve Sketching* (4.6%), 3-*Tangents* (3.4%), 4-*Maxima and Minima* (4.2%) and 5-*Rates of Change* (4.8%).
- This analysis looks at *Rates of Change*.

HSC ANALYSIS - What to expect and common pitfalls

- Rates of Change* (4.8%) as a sub-topic, is covered two areas of the syllabus: *C1 Introduction to Calculus* (standard differentiation) and *C3 Applications of Calculus* (features log, exponential and trig underlyings).
- We note questions involving integration are covered within *C4 Other Integration Applications*.
- Rates of Change* looks primarily at water flow, motion and exponential growth and decay examples. 2020 Adv 21 caused problems with the differentiation of a^x versus the normal e^x . A must review question!
- Motion featured in 2021 Adv Q26 with a substantial 5 mark allocation that proved challenging to most students.
- Thorough revision of this sub-topic must pay special attention to questions in 2016 and 2011, where students had major difficulties and mean marks were well below 50%.

Questions

1. Calculus, 2ADV C3 2022 HSC 20

A scientist is studying the growth of bacteria. The scientist models the number of bacteria, N , by the equation

$$N(t) = 200e^{0.013t},$$

where t is the number of hours after starting the experiment.

- What is the initial number of bacteria in the experiment? **(1 mark)**
- What is the number of bacteria 24 hours after starting the experiment? **(1 mark)**
- What is the rate of increase in the number of bacteria 24 hours after starting the experiment? **(2 marks)**

2. Calculus, 2ADV C3 2021 HSC 23

A population, P , which is initially 5000, varies according to the formula

$$P = 5000b^{\frac{t}{10}},$$

where b is a positive constant and t is time in years, $t \geq 0$.

The population is 1250 after 20 years.

Find the value of t , correct to one decimal place, for which the instantaneous rate of decrease is 30 people per year. **(4 marks)**

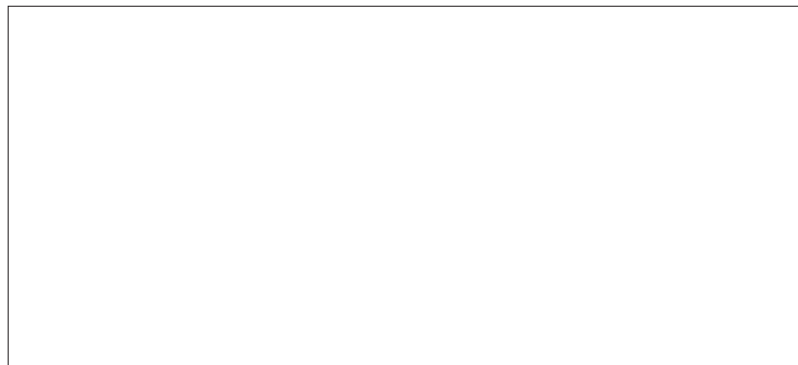
3. Calculus, 2ADV C3 2024 HSC 17

In a particular electrical circuit, the voltage V (volts) across a capacitor is given by

$$V(t) = 6.5(1 - e^{-kt}),$$

where k is a positive constant and t is the number of seconds after the circuit is switched on.

- Draw a sketch of the graph of $V(t)$, showing its behaviour as t increases. **(2 marks)**



- When $t = 1$, the voltage across the capacitor is 2.6 volts.

Find the value of k , correct to 3 decimal places. **(2 marks)**

- Find the rate at which the voltage is increasing when $t = 2$, correct to 3 decimal places. **(2 marks)**

4. Calculus, 2ADV C3 2020 HSC 21

Hot tea is poured into a cup. The temperature of tea can be modelled by $T = 25 + 70(1.5)^{-0.4t}$, where T is the temperature of the tea, in degrees Celsius, t minutes after it is poured.

- a. What is the temperature of the tea 4 minutes after it has been poured? (1 mark)
- b. At what rate is the tea cooling 4 minutes after it has been poured? (2 marks)
- c. How long after the tea is poured will it take for its temperature to reach 55°C? (3 marks)

5. Calculus, 2ADV C3 2004 HSC 5b

A particle moves along a straight line so that its displacement, x metres, from a fixed point O is given by $x = 1 + 3\cos 2t$, where t is measured in seconds.

- i. What is the initial displacement of the particle? (1 mark)
- ii. Sketch the graph of x as a function of t for $0 \leq t \leq \pi$. (2 marks)
- iii. Hence, or otherwise, find when AND where the particle first comes to rest after $t = 0$. (2 marks)
- iv. Find a time when the particle reaches its greatest magnitude of velocity. What is this velocity? (2 marks)

6. Calculus, 2ADV C3 2005 HSC 6b

A tank initially holds 3600 litres of water. The water drains from the bottom of the tank. The tank takes 60 minutes to empty.

A mathematical model predicts that the volume, V litres, of water that will remain in the tank after t minutes is given by

$$V = 3600\left(1 - \frac{t}{60}\right)^2, \text{ where } 0 \leq t \leq 60.$$

- i. What volume does the model predict will remain after ten minutes? (1 mark)
- ii. At what rate does the model predict that the water will drain from the tank after twenty minutes? (2 marks)
- iii. At what time does the model predict that the water will drain from the tank at its fastest rate? (2 marks)

7. Calculus, 2ADV C3 2021 HSC 26

A particle is shot vertically upwards from a point 100 metres above ground level.
The position of the particle, y metres above the ground after t seconds, is given by

$$y(t) = -5t^2 + 70t + 100.$$

- a. Find the maximum height above ground level reached by the particle. (2 marks)
- b. Find the velocity of the particle, in metres per second, immediately before it hits the ground, leaving your answer in the form $a\sqrt{b}$, where a and b are integers. (3 marks)

8. Calculus, 2ADV C3 2016 HSC 16b

Some yabbies are introduced into a small dam. The size of the population, y , of yabbies can be modelled by the function

$$y = \frac{200}{1 + 19e^{-0.5t}},$$

where t is the time in months after the yabbies are introduced into the dam.

- i. Show that the rate of growth of the size of the population is $\frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2}$. (2 marks)
- ii. Find the range of the function y , justifying your answer. (2 marks)
- iii. Show that the rate of growth of the size of the population can be written as $\frac{y}{400}(200 - y)$. (1 mark)
- iv. Hence, find the size of the population when it is growing at its fastest rate. (2 marks)

9. Calculus, 2ADV C3 2011 HSC 7b

The velocity of a particle moving along the x -axis is given by

$$v = 8 - 8e^{-2t},$$

where t is the time in seconds and x is the displacement in metres.

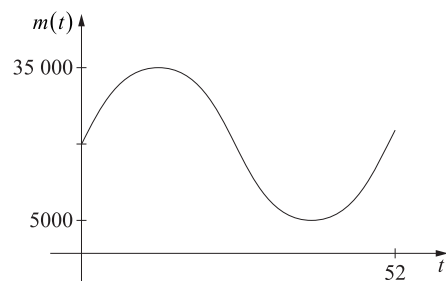
- i. Show that the particle is initially at rest. (1 mark)
- ii. Show that the acceleration of the particle is always positive. (1 mark)
- iii. Explain why the particle is moving in the positive direction for all $t > 0$. (2 marks)
- iv. As $t \rightarrow \infty$, the velocity of the particle approaches a constant.
Find the value of this constant. (1 mark)
- v. Sketch the graph of the particle's velocity as a function of time. (2 marks)

10. Trigonometry, 2ADV T3 2020 HSC 31

The population of mice on an isolated island can be modelled by the function.

$$m(t) = a \sin\left(\frac{\pi}{26}t\right) + b,$$

where t is the time in weeks and $0 \leq t \leq 52$. The population of mice reaches a maximum of 35 000 when $t = 13$ and a minimum of 5000 when $t = 39$. The graph of $m(t)$ is shown.



a. What are the values of a and b ? **(2 marks)**

b. On the same island, the population of cats can be modelled by the function

$$c(t) = -80 \cos\left(\frac{\pi}{26}(t - 10)\right) + 120$$

Consider the graph of $m(t)$ and the graph of $c(t)$.

Find the values of t , $0 \leq t \leq 52$, for which both populations are increasing. **(3 marks)**

c. Find the rate of change of the mice population when the cat population reaches a maximum. **(2 marks)**

Worked Solutions

1. Calculus, 2ADV C3 2022 HSC 20

a. $N(0) = 200e^0$
 $= 200$ bacteria

b. Find N when $t = 24$:

$$\begin{aligned} N(24) &= 200e^{0.013 \times 24} \\ &= 273.23\dots \\ &= 273 \text{ bacteria (nearest whole)} \end{aligned}$$

c. $N = 200e^{0.013t}$

$$\begin{aligned} \frac{dN}{dt} &= 0.013 \times 200e^{0.013t} \\ &= 2.6e^{0.013t} \end{aligned}$$

Find $\frac{dN}{dt}$ when $t = 24$:

$$\begin{aligned} \frac{dN}{dt} &= 2.6e^{0.013 \times 24} \\ &= 3.550\dots \\ &= 3.55 \text{ bacteria/hr (to 2 d.p.)} \end{aligned}$$

2. Calculus, 2ADV C3 2021 HSC 23

$P = 1250$ when $t = 20$

$1250 = 5000 \cdot b^{-\frac{t}{10}}$

$$b^{-2} = \frac{1}{4}$$
$$b = 2 \quad (b > 0)$$

$$P = 5000 \cdot 2^{-\frac{t}{10}}$$
$$\frac{dP}{dt} = \ln 2 \cdot -\frac{1}{10} \cdot 5000 \cdot 2^{-\frac{t}{10}}$$
$$= -500\ln 2 \cdot 2^{-\frac{t}{10}}$$

Find t when $\frac{dP}{dt} = -30$:

$$-30 = -500\ln 2 \cdot 2^{-\frac{t}{10}}$$
$$2^{-\frac{t}{10}} = \frac{3}{50\ln 2}$$

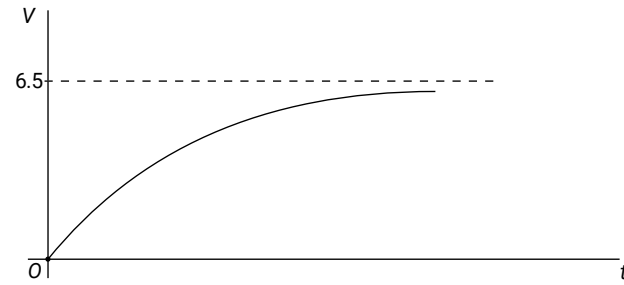
$$\ln 2^{-\frac{t}{10}} = \ln\left(\frac{3}{50\ln 2}\right)$$

$$\frac{-t}{10} = \frac{\ln\left(\frac{3}{50\ln 2}\right)}{\ln 2}$$
$$t = \frac{-10\ln\left(\frac{3}{50\ln 2}\right)}{\ln 2}$$
$$= 35.301\dots$$
$$= 35.3 \text{ years}$$

◆ Mean mark 42%.

3. Calculus, 2ADV C3 2024 HSC 17

a.



b. $V = 6.5(1 - e^{-kt})$

When $t = 1$, $V = 2.6$:

$$2.6 = 6.5(1 - e^{-k})$$

$$1 - e^{-k} = 0.4$$

$$e^{-k} = 0.6$$

$$-k = \ln(0.6)$$

$$k = 0.5108\dots$$
$$= 0.511 \text{ (3 d.p.)}$$

c. $V = 6.5 - 6.5e^{-kt}$

$$\frac{dV}{dt} = 6.5ke^{-kt}$$

Find $\frac{dV}{dt}$ when $t = 2$:

$$\frac{dV}{dt} = 6.5 \times 0.511 \times e^{-2 \times 0.511}$$
$$= 1.1953\dots$$
$$= 1.195 \text{ V/s (3 d.p.)}$$

4. Calculus, 2ADV C3 2020 HSC 21

a. $T = 25 + 70(1.5)^{-0.4 \times 4}$
 $= 61.58\dots$
 $= 61.6 \text{ (to 1 d.p.)}$

b. $\frac{dT}{dt} = 70\log_e(1.5) \times -0.4(1.5)^{-0.4t}$
 $= -28\log_e(1.5)(1.5)^{-0.4t}$

When $t = 4$,

$$\frac{dT}{dt} = -28\log_e(1.5)(1.5)^{-1.6}$$

$$= -5.934\dots$$

$$= -5.9^\circ\text{C/min (to 1 d.p.)}$$

c. Find t when $T = 55$:

$$55 = 25 + 70(1.5)^{-0.4t}$$

$$30 = 70(1.5)^{0.4t}$$

$$(1.5)^{-0.4t} = \frac{30}{70}$$

$$-0.4t\log_e(1.5) = \log_e \frac{3}{7}$$

$$-0.4t = \frac{\log_e \frac{3}{7}}{\log_e(1.5)}$$

$$\therefore t = \frac{-2.08969}{-0.4}$$

$$= 5.224\dots$$

$$= 5.2 \text{ minutes (to 1 d.p.)}$$

◆ Mean mark part (c) 44%.

5. Calculus, 2ADV C3 2004 HSC 5b

i. $x = 1 + 3\cos 2t$

When $t = 0$,

$$x = 1 + 3\cos 0$$

$$= 1 + 3$$

$$= 4$$

$\therefore$ Initial displacement is 4 m to the right of O .

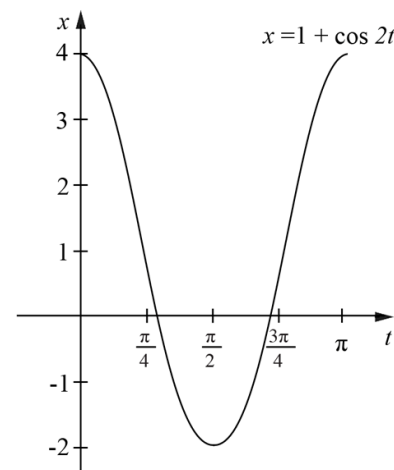
ii. Period $= \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$

Considering the range

$$-1 \leq \cos 2t \leq 1$$

$$-3 \leq 3\cos 2t \leq 3$$

$$-2 \leq 1 + 3\cos 2t \leq 4$$



iii. $x = 1 + 3\cos 2t$

$$\therefore v = -6\sin 2t$$

The particle comes to rest when $v = 0$

$$-6\sin 2t = 0$$

$$\sin 2t = 0$$

$$2t = 0, \pi, 2\pi\dots$$

$$t = 0, \frac{\pi}{2}, \pi\dots$$

∴ After $t = 0$, particle first comes to rest when

$$t = \frac{\pi}{2} \text{ seconds.}$$

When $t = \frac{\pi}{2}$,

$$\begin{aligned} x &= 1 + 3\cos 2\left(\frac{\pi}{2}\right) \\ &= 1 + 3\cos \pi \\ &= 1 + 3(-1) \\ &= -2 \end{aligned}$$

∴ Particle first comes to rest at 2 m to the left of O .

iv. $x = 1 + 3\cos 2t$

$$v = -6\sin 2t$$

$$a = -12\cos 2t$$

MAX occurs when $a = 0$

$$-12\cos 2t = 0$$

$$\cos 2t = 0$$

$$2t = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$t = \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

∴ Maximum at $t = \frac{\pi}{4}, \frac{3\pi}{4}, \dots$ seconds,

When $t = \frac{\pi}{4}$ seconds,

$$\begin{aligned} v &= -6\sin 2\left(\frac{\pi}{4}\right) \\ &= -6\sin\left(\frac{\pi}{2}\right) \\ &= -6 \end{aligned}$$

∴ Maximum is 6 m s^{-1} .

6. Calculus, 2ADV C3 2005 HSC 6b

i. $V = 3600\left(1 - \frac{t}{60}\right)^2$

When $t = 10$,

$$\begin{aligned} V &= 3600\left(1 - \frac{10}{60}\right)^2 \\ &= 3600 \times \left(\frac{5}{6}\right)^2 \\ &= 2500 \text{ L} \end{aligned}$$

ii. $V = 3600\left(1 - \frac{t}{60}\right)^2$

Using chain rule:

$$\begin{aligned} \frac{dV}{dt} &= 3600 \times 2 \times \left(1 - \frac{t}{60}\right) \times \frac{d}{dt}\left(1 - \frac{t}{60}\right) \\ &= 7200\left(1 - \frac{t}{60}\right) \times -\frac{1}{60} \\ &= -120\left(1 - \frac{t}{60}\right) \end{aligned}$$

When $t = 20$

$$\begin{aligned} \frac{dV}{dt} &= -120\left(1 - \frac{20}{60}\right) \\ &= -80 \end{aligned}$$

∴ After 20 minutes, the water will drain at 80 litres per minute.

iii. $\frac{dV}{dt} = -120\left(1 - \frac{t}{60}\right)$

$$= -120 + 2t$$

$$\frac{d^2V}{dt^2} = 2$$

Since $\frac{d^2V}{dt^2}$ is a constant, no S.P.'s

Checking limits of $0 \leq t \leq 60$

At $t = 0$,

$$\frac{dV}{dt} = -120(1 - 0) = -120 \text{ L/min}$$

At $t = 60$,

$$\frac{dV}{dt} = -120 \left(1 - \frac{60}{60} \right) = 0 \text{ L/min}$$

$\therefore$ The model predicts water will drain out the fastest when $t = 0$.

7. Calculus, 2ADV C3 2021 HSC 26

a. $y(t) = -5t^2 + 70t + 100$

$$y'(t) = -10t + 70$$

$$y''(t) = -10$$

Max height occurs when $y'(t) = 0$:

$$-10t + 70 = 0$$

$$10t = 70$$

$$t = 7$$

$$\begin{aligned} \therefore \text{Max height} &= -5(7)^2 + 70 \times 7 + 100 \\ &= 345 \text{ m} \end{aligned}$$

b. Particle hits ground when $y = 0$:

$$0 = -5t^2 + 70t + 100$$

$$0 = t^2 - 14t - 20$$

Using quadratic formula:

$$t = \frac{14 \pm \sqrt{14^2 + 4 \cdot 20}}{2}$$

$$= \frac{14 + \sqrt{276}}{2} \quad (t > 0)$$

$$= 7 + \sqrt{69}$$

$$\begin{aligned} \therefore \text{Velocity } (y'(t)) &= -10(7 + \sqrt{69}) + 70 \\ &= -10\sqrt{69} \text{ m/s} \end{aligned}$$

$$\therefore a = -10 \text{ and } b = 69$$

◆ Mean mark 38%.

8. Calculus, 2ADV C3 2016 HSC 16b

i. $y = \frac{200}{1 + 19e^{-0.5t}}$

$$\begin{aligned} \frac{dy}{dt} &= \frac{200}{(1 + 19e^{-0.5t})^2} \times \frac{d}{dt}(1 + 19e^{-0.5t}) \\ &= \frac{-200}{(1 + 19e^{-0.5t})^2} \times -0.5 \times 19e^{-0.5t} \\ &= \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2} \quad \dots \text{ as required} \end{aligned}$$

ii. When $t = 0$,

$$y = \frac{200}{1 + 19} = 10$$

$$\text{As } t \rightarrow \infty, (1 + 19e^{-0.5t}) \rightarrow 1$$

$$\therefore y \rightarrow 200$$

$$\therefore \text{Range } 10 \leq y < 200$$

◆◆◆ Mean mark (ii) 21%.

iii. $\frac{dy}{dt} = \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2}$

$$\text{Since } y = \frac{200}{1 + 19e^{-0.5t}}$$

$$\Rightarrow (1 + 19e^{-0.5t}) = \frac{200}{y}$$

$$\Rightarrow 19e^{-0.5t} = \frac{200}{y} - 1 = \frac{200 - y}{y}$$

Substituting into $\frac{dy}{dt}$:

$$\begin{aligned} \frac{dy}{dt} &= \frac{100 \left(\frac{200 - y}{y} \right)}{\left(\frac{200}{y} \right)^2} \\ &= 100 \left(\frac{200 - y}{y} \right) \times \frac{y^2}{200^2} \\ &= \frac{y}{400} (200 - y) \quad \dots \text{ as required} \end{aligned}$$

iv. $\frac{dy}{dt} = -\frac{y^2}{400} + \frac{y}{2}$

Sketching the parabola:

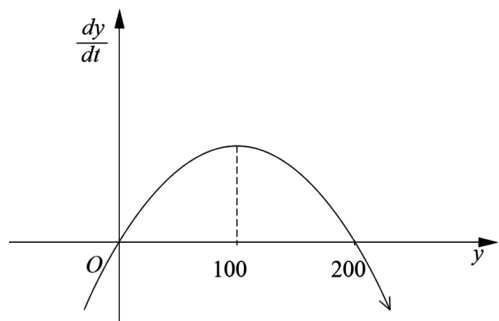
$$\frac{-y^2}{400} + \frac{y}{2} = 0$$

$$-y^2 + 200y = 0$$

$$y(200 - y) = 0$$

◆◆◆ Mean mark (iii) 18%.

◆◆◆ Mean mark (iv) 14%.



$\therefore$ Maximum $\frac{dy}{dt}$ occurs when $y = 100$.

9. Calculus, 2ADV C3 2011 HSC 7b

i. Initial velocity when $t = 0$

$$\begin{aligned} v &= 8 - 8e^0 \\ &= 0 \text{ m/s} \end{aligned}$$

$\therefore$ Particle is initially at rest.

ii. $a = \frac{d}{dt}(v) = -2 \times -8e^{-2t} = 16e^{-2t}$

Since $e^{-2t} = \frac{1}{e^{2t}} > 0$ for all t .

$$\Rightarrow a = 16e^{-2t} = \frac{16}{e^{2t}} > 0 \text{ for all } t.$$

$\therefore$ Acceleration is positive for all $t > 0$.

iii. Since the particle is initially at rest, and ALWAYS has a positive acceleration.

$\therefore$ It moves in a positive direction for all t .

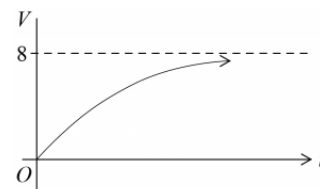
iv. As $t \rightarrow \infty$, $e^{-2t} = \frac{1}{e^{2t}} \rightarrow 0$

$$\Rightarrow \frac{8}{e^{2t}} \rightarrow 0 \text{ and}$$

$$\Rightarrow v = 8 - \frac{8}{e^{2t}} \rightarrow 8 \text{ m/s}$$

$\therefore$ As $t \rightarrow \infty$, velocity approaches 8 m/s.

v.



MARKER'S COMMENT: Students whose working showed e^{-2t} as $\frac{1}{e^{2t}}$, tended to score highly in this question.

◆◆◆ Mean mark 22%
COMMENT: Students found part (iii) the most challenging part of this question by far.

IMPORTANT: Use previous parts to inform this diagram. i.e. clearly show velocity was zero at $t = 0$ and the asymptote at $v = 8$.

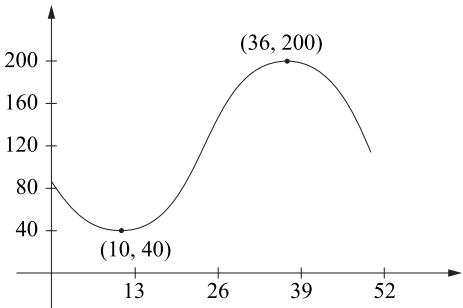
10. Trigonometry, 2ADV T3 2020 HSC 31

a. $b = \frac{35\,000 + 5000}{2}$
 $= 20\,000$

$a =$ amplitude of sin graph
 $= 35\,000 - 20\,000$
 $= 15\,000$

b. By inspection of the $m(t)$ graph
 $m'(t) > 0$ when $0 \leq t < 13$ and $39 < t \leq 52$
Sketch $c(t)$:

♦♦ Mean mark part (b) 30%.



Minimum $(\cos 0)$ when $t = 10$
Maximum $(\cos \pi)$ when $t = 36$
 $\therefore c'(t) > 0$ when $10 < t < 36$
 $\therefore$ Both populations are increasing when $10 < t < 13$

c. $c(t)$ maximum when $t = 36$

♦♦♦ Mean mark part (c) 27%.

$$m(t) = 15\,000 \sin\left(\frac{\pi}{26}t\right) + 20\,000$$
$$m'(t) = \frac{15\,000\pi}{26} \cos\left(\frac{\pi}{26}t\right)$$
$$m'(36) = \frac{15\,000\pi}{26} \cdot \cos\left(\frac{36\pi}{26}\right)$$
$$= -642.7$$

$\therefore$ Mice population is decreasing at 643 mice per week.